

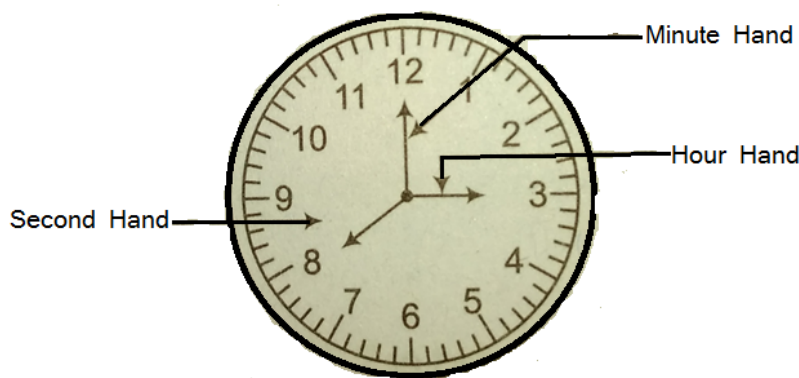
Clock and Calender

Clock

A clock is an instrument used for indicating and maintaining the time. It is an electronic device that presents the duration of an hour, minute and second.

Different types of question covered in this chapter are as follows.

- **Angle Between the Hands of the clock**
- **Position of Hands of the clock**
- **Faulty clocks**
- **Time Gained or lost by the clock**



The clock represents two things. i.e., minutes an hour. A minute is a unit of time equal to $1/60^{\text{th}}$ of an hour or 60 s i.e, **1 min = 60 s**.

Clocks Concepts :

- The dial of the clock is circular in shape and was divided into 60 equal minute spaces
- 60-minute spaces trace an angle of **360°** . Therefore, 1minute space traverses an angle of 6°
- **In 1 hour, Minute hand traverses 60-minute space or 360°** , Hour hand traverses 5-minute space or 30°
- The hands of the clock are **perpendicular** in 15-minute spaces apart
- The hands of the clock are in straight line and **opposite** to each other in **30-minute** spaces apart.
- The hands of the clock are in straight line when they coincide or opposite to each other.

- The hands of the clock are perpendicular to **each other for 22 times in 12 hours** and for **44 times** in a day.
- The hands of the clock are **opposite** to each other for **11 times in 12 hours** and 22 times in a day.
- The minute hand gain **55 minutes** over hour hand per hour.

Hence x minute space to be gained by minute hand over hour hand can be calculated as $x \cdot (60/55)$ or $x \cdot (12/11)$

Ex: At what time between 2'O clock and 3'O clock the hands of the clock Ware opposite to each other.

1. 34(6/11) past 2'Oclock

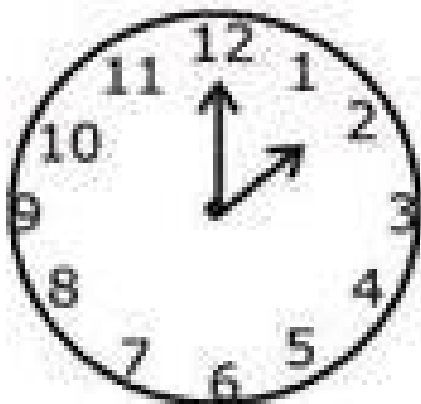
2. 43(7/11) past 2'Oclock

3. 56(8/11) past 2'Oclock

4. 64(9/11)past 2'Oclock

Solution:

At 2'O clock the minute hand will be at 12 as shown below



The minute's hand to coincide with the hour hand it should trace at first 10 minute spaces and then the hands of the clocks to be opposite to each other minute hand should trace 30-minute spaces i.e. totally it should gain $10+30=40$ minute spaces to be opposite to that of the hour hand

We know that,

Minute hand gains 55-minute spaces over hour hand in 1 hour

Therefore, Minute hand gain 40-minute spaces over hour hand in $40 \times (60/55) = 43(7/11)$

Hence the hand of the clock & minute will be opposite to each other at 43(7/11) past 2'Oclock

Therefore, **the Correct option is 2'**

When the clock is too fast, too slow

- If a clock or watch indicates 6 hr 10 min when the correct time is 6, it is said that the clock is 10 min too fast. If it indicates 6. 40 when the correct time is 7, it is said to be 20 min too slow.
Now let us have an example based on this concept

Ex: My watch, which gains uniformly, is 2 min, & show at noon on Sunday, and is 4 min 48 seconds fast at 2 p.m on the following Sunday when was it correct?

Solution: From Sunday noon to the following Sunday at 2 p.m there are 7 days 2 hours or 170 hours.

The watch gains $2 + 4\frac{4}{5}$ min in 170 hrs.

Therefore, the watch gains 2 min in $(2 \times 170 \times \frac{5}{34})$ hrs i.e., 50 hours

Now 50 hours = 2 days 2 hrs.

Therefore, 2 days 2 hours from Sunday noon = 2 p.m on Tuesday.

Problems On angles

Before we actually start solving problems on angles, we need to know a couple of basic facts clear:

- **Speed of the hour hand** = 0.5 degrees per minute (dpm)
- **Speed of the minute hand** = 6 dpm
- At 'n' o' clock, the angle of the hour hand from the vertical is $30n$

Example: 1 *What is the angle between the hands of the clock at 7:20*

Solution:

At 7 o' clock, the hour hand is at 210 degrees from the vertical.

In 20 minutes,

Hour hand = $210 + 20 \times (0.5) = 210 + 10 = 220$ {The hour hand moves at 0.5 dpm}

Minute hand = $20 \times (6) = 120$ {The minute hand moves at 6 dpm}

Difference or angle between the hands = $220 - 120 = 100$ degrees

Example :2 *Find the reflex angle between the hands of a clock at 05.30?*

Solution:

Angle between X and Y = $| (X \times 30) - ((Y \times 11) / 2) |$

Angle between hands at 5:30

Step 1: $X=5$, $Y=30$

Step 2: $5 \times 30 = 150$

Step 3: $(30 \times 11) / 2 = 165$

Step 4: $165 - 150 = 15$

Thus, the angle between hands at 5:30 is 15 degrees.

Example : 3 *At what time 3&4'o clock in the hands of clock together.*

Solution:

Approximately we know at 03:15 hands of the clock together

So $15 \times 60 / 55 = 16.36$ min

Problems on incorrect clocks

Such sort of problem arises when a clock runs faster or slower than expected pace. When solving these problems it is best to keep track of the correct clock.

Example: *A watch gains 5 seconds in 3 minutes and was set right at 8 AM. What time will it show at 10 PM on the same day?*

Solution:

The watch gains 5 seconds in 3 minutes = 100 seconds in 1 hour.

From 8 AM to 10 PM on the same day, time passed is 14 hours.

In 14 hours, the watch would have gained 1400 seconds or 23 minutes 20 seconds.

So, when the correct time is 10 PM, the watch would show 10 : 23 : 20 PM

Important Points to remember:

- Two right angles per hour (Right angle = 90, Straight angle=180)
- Forty four right angles per day
- Between every two hours the hands of the clock coincide with each other for one time except between 11, 12 and 12, 1. In a day they coincide for

22 times.

- Between every two hours they are perpendicular to each other two times except between 2, 3 and 3, 4 and 8, 9 and 9, 10. In a day they will be perpendicular for 44 times.
- Between every two hours they will be opposite to each other one time except between 5, 6 and 6, 7. In a day they will be opposite for 22 times.

Clocks & Calendars

Instructions

For the following questions answer them individually

Question 1

Find the mirror image of the clock when the time is 01:40

- A** 11:20
- B** 10:22
- C** 10:20
- D** 11:22

Answer: C

Explanation:

We need to subtract from 12:00 or 11:60 to get mirror image time

Mirror image of 01:40 = 11:60 - 01:40 = 10:20

Question 2

Find the mirror image of the clock when the time is 02:40

- A** 09:20
- B** 10:22
- C** 09:25
- D** 09:22

Answer: A

Explanation:

We need to subtract from 12:00 or 11:60 to get mirror image time

Mirror image of 02:40 = 11:60 - 02:40 = 09:20

Question 3

At what time between 2 O'Clock and 3 O'Clock, will the minute hand and hour hand of the clock be exactly opposite to each other?

- A** 02 : 46
- B** 02 : 47⁶₁₁
- C** 02 : 43⁷₁₁
- D** 02 : 47

Answer: C

Explanation:

Angle between two hands = $\frac{11}{2} M - 30H$ where H is hours and M is Minutes

Here, H = 2 and Angle = 180°

$$180^\circ = \frac{11}{2} M - 30 \times 2$$

$$\begin{aligned}
 & \Rightarrow 2M = 240 \\
 & \Rightarrow M = \frac{240}{2} = 120 \\
 & \Rightarrow M = 120 = 2 \times 60
 \end{aligned}$$

Hence, the required time = 02 : 43 11

Question 4

At what time between 12 O’Clock and 1 O’Clock, will the minute hand and hour hand of the clock make right angles?

- A 12 : 46
- B 12 : 15 11
- C 12 : 17 11
- D None of these

Answer: D

Explanation:

Angle between two hands = $2M - 30H$ where H is hours and M is Minutes

Here, H = 12 and Angle = 90°

$$90^\circ = 2M - 30 \times 12$$

$$\Rightarrow 2M = 450$$

$$\Rightarrow M = \frac{450}{2} = 225$$

Hence, the required time = 12 : 21 11 which is 01 : 21 11

Hence, Between 12 O’Clock and 1 O’Clock, there is no right angle.

Question 5

At what time between 2 O’Clock and 3 O’Clock, will the minute hand and hour hand of the clock make right angles?

- A 02 : 32
- B 02 : 29 11
- C 02 : 27 11
- D 02 : 36

Answer: C

Explanation:

$$11$$

Angle between two hands = $2M - 30H$ where H is hours and M is Minutes

Here, H = 2 and Angle = 90°

$$11$$

$$90^\circ = 2M - 30 \times 2$$

$$11$$

$$\Rightarrow 2M = 150$$

$$300$$

$$3$$

$$\Rightarrow M = \frac{150}{2} = 75$$

$$3$$

Hence, the required time = 02 : 27 11

Question 6

At what time between 4 O'Clock and 5 O'Clock, will the minute hand and hour hand of the clock make right angles?

A 04 : 36

$$2$$

B 04 : 38 11

$$4$$

C 04 : 42 11

D 04 : 39

Answer: B

Explanation:

$$11$$

Angle between two hands = $2M - 30H$ where H is hours and M is Minutes

Here, H = 4 and Angle = 90°

$$11$$

$$90^\circ = 2M - 30 \times 4$$

$$11$$

$$\Rightarrow 2M = 210$$

$$420$$

$$2$$

$$\Rightarrow M = \frac{210}{2} = 105$$

$$2$$

Hence, the required time = 04 : 38 11

Question 7

A clock shows 7 O'clock in the morning. By how much angle will the hours hand rotate when the clock shows 9 O'clock in the morning.

A 40°

B 60°

C 45°

D 90°

Answer: B

Explanation:

In 12 hours, the hand turns 360° .

Here, the difference between time = 2 hours

$$\frac{360}{12}$$

Then, Required angle = $12 \times 2 = 60^\circ$

Question 8

A clock shows 6:00 in the morning. By how much angle will the hours hand rotate when the clock shows 12:00 in the noon?

A 120°

B 160°

C 180°

D 150°

Answer: C

Explanation:

In 12 hours, the hand turns 360° .

Here, the difference between time = 6 hours

$$\frac{360}{12}$$

Then, Required angle = $12 \times 6 = 180^\circ$

Question 9

Find the angle between the hands of the clock when the time is 10:30

A 160°

B 120°

C 180°

D 135°

Answer: D

Explanation:

Required angle = $30H - \frac{11}{2}M$ where H is hours and M is Minutes

Here, H = 10 and M = 30

Hence, Required angle = $30 \times 10 - \frac{11}{2} \times 30$

$$= 300 - 165 = 135^\circ$$

Question 10

Check which of the following years are leap years.

A 1800

B 1345

C 1678

D none of these

Answer: D

Explanation:

1800 is century year so it must be divisible by 400 to be a leap year.

1345, 1678 are not divisible by 4,

So the answer is option D.

Question 11

March 10, 2018 is Saturday. What day of the week lies on March 10, 2019 ?

A Thursday

B Friday

C Saturday

D Sunday

Answer: D

Explanation:

The year 2018 is non-leap year, so it has 365 days, $365 = 52 \text{ weeks} + 1 \text{ odd day}$

Hence one day beyond Saturday is Sunday.

So the answer is option D.

Question 12

Today is Saturday, After 43 days it will be ?

A Wednesday

B Friday

C Sunday

D Tuesday

Answer: C

Explanation:

Today is Saturday.

Each day of the week is repeated after every 7 days.

Hence, after 42 days it will be Saturday.

So after 43 days, it will be Sunday.

So the answer is option C.

Question 13

Sridevi was born on 13-8-1963 and died on 24-2-2018. Find her exact age ?

A 54 years 6 months 12 days

B 54 years 5 months 22 days

C 54 years 5 months 12 days

D 54 years 6 months 22 days

Answer: A

Explanation:

13-8-1963 to 12-8-2017 = 54 years

13-8-2017 to 12-2-2018 = 6 months

13-2-2018 to 24-2-2018 = 12 days

So the answer is option A.

Question 14

Find the average number of days per month in the year 1994 ?

A 30.4266

B 30.4166

C 30.4066

D 30.3966

Answer: B

Explanation:

1994 has 365 days

average days per month = $\frac{365}{12} = 30.4166$

So the answer is option B.

Question 15

If 1st january of 1996 was Monday, then How many tuesdays did 1996 have ?

A 53

B 52

C 51

D cannot be determined

Answer: A

Explanation:

1996 has 366 days = 52 weeks + 2days

If 1st jan 1996 was monday, it would have 53 mondays and 53 tuesdays

So the answer is option A.

Clock Questions and Answers

Q 1. In a clock displayed at a wall, the time displayed is 9 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 3 o'clock in the afternoon?

1. 120 degree
2. 160 degree
3. 180 degree
4. 260 degree
5. 150 degree

Answer: (3) 180 degree

Solution:

There is a difference of 6 hours between 9 o'clock in the morning and 3 o'clock in the afternoon
Degrees with which the hour hand clock will rotate = $(360/12) \times 6 = 180$ degree

Q 2. How many minutes are lost by a clock per day, if its hands coincide every 54 minutes?

1. $35 \frac{5}{11}$ minutes
2. $205 \frac{5}{11}$ minutes
3. $305 \frac{5}{11}$ minutes
4. $25 \frac{5}{11}$ minutes
5. Cannot be determined

Answer: (3) $305 \frac{5}{11}$ minutes

Solution:

In 60 minutes, there a total of 55 minute spaces which are covered.

60 minute spaces are covered in $(\frac{60}{55} \times 60)$ minutes = $65 \frac{5}{11}$ minutes

Loss in time in 54 minutes = $(65 \frac{5}{11}) - 54 = 11 \frac{5}{11}$ minutes

Loss of minutes in 24 hours = $11 \frac{5}{11} \times \frac{1}{54} \times 24 \times 60 = 305 \frac{5}{11}$ minutes

Q 3. At what time between 6 a.m. and 6:25 a.m. will the hands of a clock be at right angles?

1. 44 minutes
2. $43 \frac{7}{11}$ minutes
3. $42 \frac{7}{11}$ minutes
4. 45 minutes
5. 42 minutes

Answer: (2) $43 \frac{7}{11}$ minutes

Solution:

Between 6 a.m. and 6:25 a.m., the hands are at 20 minutes space apart

So, to make a right angle between the hands of the clock, 20 more minutes shall be required, which is = $20+20 = 40$ spaces

55 minute spaces are gained in 60 minutes

Thus, 40 minute spaces are gained in $(\frac{60}{55} \times 40) = 43 \frac{7}{11}$ minutes

Q 4. At what time between 7 o'clock and 8 o'clock will the hands of a wall clock point in opposite directions?

1. $54 \frac{6}{11}$ minutes past 7
2. $53 \frac{6}{11}$ minutes past 7
3. $52 \frac{6}{11}$ minutes past 7
4. Cannot be Determined
5. None of the above

Answer: (1) $54 \frac{6}{11}$ minutes past 7

Solution:

At 7 o'clock the hands of the watch are 20 spaces apart

For the hands of the clocks to be in opposite direction, the spaces between them must be 30 minutes
50 minute spaces will have to be gained by the minute hand

55 minute spaces are gained in 60 minutes

Thus, 50 minute spaces are gained in $(\frac{60}{55} \times 50) = 54 \frac{6}{11}$ minutes

Thus, the required time is $54 \frac{6}{11}$ minutes past 7 o'clock

Q 5. Between 4 o'clock and 5 o'clock in the morning, at what time will the hands of a wall clock be in the same straight line? But these lines are not together, i.e., not overlapping each other.

1. $6 \frac{5}{11}$ minutes past 4
2. $5 \frac{5}{11}$ minutes past 4
3. $5 \frac{4}{11}$ minutes past 4
4. $4 \frac{4}{11}$ minutes past 4
5. None of the above

Answer: (2) $5 \frac{5}{11}$ minutes past 4

Solution:

If the two hands of the clock are in a straight line but not together, then they are 30 spaces apart

At 4 o'clock, they are 25 minute spaces apart

Thus, the minutes hand will have to gain only 5 minute spaces

We know, 55 minute spaces are gained in 60 minutes

So, 5 minute spaces are gained in $(\frac{60}{55} \times 5) = \frac{60}{11}$ minutes = $5 \frac{5}{11}$ minutes

So, the two hands will be in a straight line at $5 \frac{5}{11}$ minutes past 4

Q 6. What is the angle between the hour hand and the minute hand when the time is 7:30?

1. 32°
2. 43°
3. 54°
4. 45°
5. 23°

Answer: (4) 45°

Solution:

Angle traced by hour hand in $15/2$ hours = $(\frac{360}{12} \times \frac{15}{2})^\circ = 225$

Angle traced by minute hand in 30 minutes = $\frac{360}{60} \times 30^\circ = 180$

Thus, the angle between the hour and minute hand when the time is 7:30 = $(225-180)^\circ = 45^\circ$

Q 7. At what time between 3 o'clock and 4 o'clock, both the minute and hour hand will coincide with each other? Find the answer in minutes.

1. $16 \frac{7}{11}$ minutes past 3
2. $14 \frac{4}{11}$ minutes past 3
3. $13 \frac{2}{11}$ minutes past 3
4. $16 \frac{4}{21}$ minutes past 3
5. $16 \frac{4}{11}$ minutes past 3

Answer: (5) $16 \frac{4}{11}$ minutes past 3

Solution:

The minute hand is 15 minute spaces apart from the hour hand at 3 o'clock

To coincide with each other it must gain 15 minute spaces

We know, 55 minute spaces are gained in 60 minutes

So, spaces gained in 15 minutes = $\frac{60}{55} \times 15 = 16 \frac{4}{11}$ minutes

Thus, both the hands will coincide at $16 \frac{4}{11}$ minutes past 3

Q 8. If the second hand moves 3600 times, then how many degrees will the minute hand move at the same time?

1. 60°
2. 25°
3. 30°
4. 45°
5. 90°

Answer: (1) 60°

Solution:

1 minute = 60 seconds

Thus, degree at which the minute hand shall move = $3600/60 = 60^\circ$

Q 9. At 5 o'clock a clock ticks 5 times. The time gap between the first and last tick is 40 seconds. How long does the clock tick when it is 12 o'clock in the clock?

1. 60 seconds
2. 95 seconds
3. 110 seconds
4. 105 seconds
5. 75 seconds

Answer: (3) 110 seconds

Solution:

If at 7 o'clock, the clock ticks 5 times, then it means that there were 4 time intervals.

Thus, the time duration of each interval = $40/4 = 10$ seconds

At 12 o'clock, the clock will tick 12 times and the number of intervals shall be 11

So, the total time for which the clock ticks = $11 \times 10 = 110$ seconds

Q 10. At what angle will the hour hand be turned through at 10 minutes past 5, if the clock is started at noon?

1. 160°
2. 122°
3. 132°
4. 150°
5. 155°

Answer: (5) 155°

Solution:

Angle traced by the hour hand in 12 hours = 360°

We know, 5 hours 10 minutes = $31/6$ hours

So, angle traced by the hour hand in 5 hours 10 minutes = $(\frac{360}{12} \times \frac{31}{6}) = 155$

Thus, 155° angle is formed when the hour hand be turned through at 10 minutes past 5 and the clock is started at noon

Q 11. What is the ratio of 15 minutes to 1 hour?

1. 1:2
2. 3:5
3. 5:20
4. 3:14
5. 5:21

Answer: (3) 5:20

Solution:

1 hour = 60 minutes

So, the ratio of 15 minutes to 1 hour = $15:60 = 5:20$

Q 12. What is the ratio of 42 minutes to 2 hours?

1. 7:20
2. 14:15
3. 1:2
4. 3:5
5. 6:7

Answer: (1) 7:20

Solution:

1 hour = 60 minutes

2 hours = $60 \times 2 = 120$ minutes

Thus, the ratio of 42 minutes to 2 hours = $42:120 = 7:20$

Q 13. Vinit bought a new wall clock on Monday and set the correct time as 10 a.m. on it and fixed it on a wall. The same clock loses 15 minutes in 24 hours. What will be the true time if the clock indicates 4 a.m. on the following Sunday?

1. 6:00 a.m.
2. 5:02 a.m.
3. 4:44 a.m.
4. 5:12 a.m.
5. 6:43 a.m.

Answer: (4) 5:12 a.m.

Solution:

From the time the clock was set to the final time taken, the total hours = 114 (From Monday 10 a.m. to Sunday 4 a.m.)

If the clock is losing 15 minutes in one day, it means that the rest 23 hours and 245 minutes is correctly displayed.

Which means, $95/4$ hours of the incorrect time = 24 hours of the correct clock

So, $(24 \times \frac{4}{95}) \times 114 = 115 \frac{19}{95}$ hours or 115.2 hours

Therefore, 115 hours 12 minutes of the correct clock

So, the true time will be 5:12 a.m.

Q 14. What is the ratio of 3 hours to 54 minutes?

1. 5:2
2. 11:9
3. 2:1
4. 9:4
5. 10:3

Answer: (5) 10:3

Solution:

1 hour = 60 minutes

3 hours = $60 \times 3 = 180$ minutes

So, the ratio of 3 hours to 54 minutes = $180:54 = 10:3$

Q 15. In a wristwatch, at what time between 1 o'clock and 2 o'clock, will the minutes and hours hands of the watch make an angle of 180° ?

1. $35 \frac{2}{11}$ minutes past 1
2. $38 \frac{2}{11}$ minutes past 1
3. $39 \frac{2}{11}$ minutes past 1
4. $36 \frac{2}{11}$ minutes past 1

5. $37\frac{2}{11}$ minutes past 1

Answer: (2) $38\frac{2}{11}$ minutes past 1

Solution:

Let us assume that when the minute's hand is at 7, the hands are going to make an 180° angle

So, it can be said that it will contain 7 equal parts, with the degree of each part = 30°

So, the total angle = $30^\circ \times 7 = 210^\circ$

If $11\frac{1}{2}^\circ$ is gained in a minute

Then, 210° gain will be = $\frac{2}{11} \times 210 = 420/11$

$\Rightarrow 38\frac{2}{11}$ minutes past 1

Q 16. What shall be the ratio of 18 minutes to 9 hours?

1. 2:29
2. 1:30
3. 30:1
4. 29:2
5. 3:4

Answer: (2) 1:30

Solution:

1 hour = 60 minutes

9 hours = $60 \times 9 = 540$ minutes

So, the ratio of 18 minutes to 9 hours = $18:540 = 1:30$

Clock and Calender

TYPE-I

1. If 9th of the month falls on the day preceding Sunday, on what day will 1st of the month fall?

(1) Friday (2) Saturday
(3) Sunday (4) Monday

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (First Sitting))

2. Anil reached a place on Friday. He came to know that he was three days earlier than the scheduled day. If he had reached there on the following Sunday, how many days late/early he would have been?

(1) One day earlier
(2) One day late
(3) Two days late
(4) Two days earlier

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Second Sitting))

3. If day before yesterday was Friday, what will be the third day after the day – after- tomorrow?

(1) Thursday (2) Friday
(3) Saturday (4) Sunday

(SSC CPO Sub-Inspector Exam. 09.11.2008)

4. If the day before yesterday was Thursday, when will Sunday be?

(1) Tomorrow
(2) Day after tomorrow
(3) Today
(4) Two days after today

(SSC CPO Sub-Inspector Exam. 06.09.2009)

5. If the day before yesterday was Sunday, what day will it be three days after the day after tomorrow?

(1) Sunday (2) Monday
(3) Wednesday (4) Saturday

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (First Sitting))

6. If the day after tomorrow is Sunday, what day was tomorrow's day before yesterday?

(1) Friday (2) Thursday
(3) Monday (4) Tuesday

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (Second Sitting))

7. Suresh was born on 4th October 1999. Shashikanth was born 6 days before Suresh. The Independence Day of that year fell on Sunday. Which day was Shashikanth born?

(1) Tuesday (2) Wednesday
(3) Monday (4) Sunday

(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (First Sitting))

8. Reaching a place of appointment on Friday, I found that I was two days earlier than the scheduled day. If I had reached on the following Wednesday, how many days late would I have been?

(1) One day (2) Two days
(3) Three days (4) Four days

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (IInd Sitting) (Middle Zone, Allahabad))

9. If the 23rd of a month is a Sunday, what day it would have been two weeks and four more days earlier?

(1) Monday (2) Tuesday
(3) Wednesday (4) Thursday

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (IInd Sitting))

10. If 15th June falls 3 days after tomorrow, that is Friday, on what day will the last of the month fall?

(1) Monday (2) Tuesday
(3) Wednesday (4) Thursday

(SSC Combined Matric Level (PRE) Exam. 27.05.2001 (IInd Sitting) (East Zone))

11. If the day after tomorrow is Tuesday what day will two days after the day after tomorrow be?

(1) Monday (2) Wednesday
(3) Saturday (4) Thursday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (Ist Sitting) (Eastern Zone, Guwahati))

12. If two days before yesterday was Friday, what day will be day after tomorrow?

(1) Monday (2) Sunday
(3) Saturday (4) Wednesday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (Ist Sitting) (Eastern Zone, Guwahati))

13. The day before yesterday was Sunday. What will be the day after tomorrow?

(1) Monday (2) Thursday
(3) Friday (4) Saturday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (Eastern Zone, Guwahati))

14. If two days after day after tomorrow is Sunday, what day was the day before yesterday?

(1) Wednesday (2) Tuesday
(3) Monday (4) Sunday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (Eastern Zone, Guwahati))

15. If day before yesterday was Tuesday, the day after tomorrow will be

(1) Monday (2) Wednesday
(3) Friday (4) Saturday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (Ist Sitting) (North Zone, Delhi))

16. If three days after today will be Tuesday, what day was four days before yesterday?

(1) Tuesday (2) Sunday
(3) Monday (4) Wednesday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (Ist Sitting) (North Zone, Delhi))

17. If day before yesterday was Thursday, what day will be four days after tomorrow?

(1) Monday (2) Thursday
(3) Sunday (4) Wednesday

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (North Zone Delhi))

18. If the day that will come two days after tomorrow is Thursday, what day of the week was three days before yesterday?

- (1) Monday (2) Tuesday
(3) Wednesday (4) Thursday
SSC Combined Matric Level (Pre)
Exam. 05.05.2002 (IInd Sitting)
(North Zone Delhi)
- 19.** If three days before yesterday was Wednesday, what will be two days after tomorrow?
(1) Wednesday (2) Monday
(3) Friday (4) Tuesday
SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (Ist Sitting)
- 20.** If Friday is the first day of a non-leap year, what day would the last day of the year be?
(1) Friday (2) Sunday
(3) Monday (4) Tuesday
SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (Ist Sitting)
- 21.** If day after tomorrow is Saturday what day was three days before yesterday?
(1) Thursday (2) Monday
(3) Saturday (4) Sunday
SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting)
- 22.** Day after tomorrow is Kiran's birthday. On the same day next week falls 'Shivratri'. Today is Monday. What will be the day after 'Shivratri'?
(1) Wednesday (2) Thursday
(3) Friday (4) Saturday
SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)
- 23.** If the day before yesterday was Thursday, when will Sunday be?
(1) Day after tomorrow
(2) Today (3) Tomorrow
(4) Two days after today
SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (Ist Sitting) (East Zone)
- 24.** If the day after tomorrow is Friday, what day will third day after the tomorrow be?
(1) Saturday (2) Monday
(3) Sunday (4) Friday
SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (IInd Sitting) (Central Zone)
- 25.** Mohini went to movies nine days ago. She goes to the movies only on Thursday. What day of the week is today?
(1) Saturday (2) Thursday
(3) Sunday (4) Tuesday
(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 27.11.2010)

- 26.** If the day before yesterday was Thursday, when will Sunday be?
(1) Today
(2) Two days after today
(3) Tomorrow
(4) Day after Tomorrow
(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (Ist sitting))
- 27.** If day before yesterday was Wednesday, when will Sunday be?
(1) 3 days after today
(2) Tomorrow
(3) Today
(4) Day after tomorrow
(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (IInd sitting))
- 28.** If the day before yesterday was Friday, what day will two days after the day after tomorrow be?
(1) Saturday (2) Thursday
(2) Friday (4) Sunday
(SSC Stenographer Grade 'C' & 'D'
Exam. 09.01.2011)
- 29.** Ann is 300 days older than Varun and Sandeep is 50 weeks older than Ann. If Sandeep was born on Tuesday, on which day was Varun born?
(1) Monday (2) Thursday
(3) Tuesday (4) Friday
(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Patna Region :
Ist Sitting)
- 30.** If 1st March is Saturday, then 1st April will be
(1) Sunday (2) Monday
(3) Tuesday (4) Thursday
(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Ist Sitting
TF No. 333 LO 2)
- 31.** If it is Saturday on 27th September, what day will it be on 27th October of the same year?
(1) Thursday (2) Sunday
(3) Friday (4) Monday
(SSC CGL Tier-I (CBE)
Exam. 31.08.2016 (Ist Sitting))
- 32.** If the third day of a month is Tuesday, which of the following would be the 25th day of that month?
(1) Tuesday (2) Monday
(3) Wednesday (4) Sunday
(SSC CGL Tier-I (CBE)
Exam. 31.08.2016 (IInd Sitting))

TYPE-II

- 1.** Mrs. Susheela celebrated her wedding anniversary on Tuesday, 30th September 1997. When will she celebrate her next wedding anniversary on the same day?
(1) 30 September 2003
(2) 30 September 2004
(3) 30 September 2002
(4) 30 October 2003
(SSC Combined Graduate Level
Prelim Exam.19.06.2011
(First Sitting))
- 2.** If John celebrated his victory day on Tuesday, 5th January 1965, when will he celebrate his next victory day on the same day?
(1) 5th January 1970
(2) 5th January 1971
(3) 5th January 1973
(4) 5th January 1974
(SSC Combined Graduate Level Prelim
Exam.19.06.2011(Second Sitting))
- 3.** In the year 1996, the Republic day was celebrated on Friday. On which day was the Independence day celebrated in the year 2000?
(1) Tuesday (2) Monday
(3) Friday (4) Saturday
(SSC Combined Graduate Level Tier-1
Exam. 26.06.2011(Second Sitting))
- 4.** Hari remembers that his father's birthday is between 13th and 16th of June, whereas his sister remembers that their father's birthday is between 14th and 18th of June. On which day is their father's birthday, which both agree?
(1) 14th June (2) 15th June
(3) 16th June (4) 17th June
(SSC CPO (SI, ASIn & Intelligence Officer)
Exam. 28.08.2011 (Paper-I))
- 5.** 5th of a month falls two days after Monday. What day of the month will precede 19th of it?
(1) Wednesday (2) Thursday
(3) Tuesday (4) Monday
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (Ist Sitting))
- 6.** The Independence day in 1988 was celebrated on a Wednesday. On what day was it celebrated in the year 1989?

SCHEDULED DAY/DATE/TIME

- (1) Monday (2) Tuesday
(3) Friday (4) Thursday
(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (1st Sitting))
7. Tell the number of days from 26 January 2006 to 23 September 2006 (including both dates).
(1) 214 (2) 241
(3) 249 (4) 251
(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (1st Sitting)
(Eastern Zone, Guwahati))
8. Today is Friday. On the last Monday the date was 29th December, 1975. The date today is
(1) 28 December, 1975
(2) 3rd January, 1976
(3) 2nd January, 1976
(4) 2nd January, 1975
(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (1st Sitting)
(Eastern Zone, Guwahati))
9. If the third Friday of a month is 16th what date is the fourth Tuesday of that month?
(1) 20th (2) 22nd
(3) 27th (4) 29th
(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (IInd Sitting)
(Eastern Zone, Guwahati))
10. If Friday falls on 15th of September 2000, what will be the day of 15th of September 2001?
(1) Friday (2) Saturday
(3) Thursday (4) Sunday
(SSC Combined Matric Level (Pre)
Exam. 05.05.2002 (IInd Sitting)
(North Zone Delhi))
11. Find out the day from the problem given. If the 1st of November falls on Monday, what day will the 25th of November be?
(1) Tuesday (2) Thursday
(3) Wednesday (4) Friday
(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (1st Sitting))
12. Saturday comes after 3 days on 4th of a month. Find out the day on 27th of that month.
(1) Monday (2) Thursday
(3) Friday (4) Saturday
(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting))
13. If the 2nd of a month falls on Sunday, what day will the 31st of that month be?
(1) Tuesday (2) Saturday
(3) Friday (4) Monday
(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting))

14. Given that 25th February 2008 is Monday, what day is 2nd March of 2008?
(1) Tuesday (2) Saturday
(3) Sunday (4) Monday
(SSC Level Data Entry Operator & LDC Exam. 21.10.2012 (IInd Sitting))
15. If 8th of April falls on Monday, what would be the 30th day of that month?
(1) Sunday (2) Monday
(3) Tuesday (4) Wednesday
(SSC Constable (GD)
Exam. 12.05.2013 1st Sitting)
16. If two days back it was 9th November 2014 and it was a Sunday, then tomorrow will be
(1) 12th November 2014, Wednesday
(2) 12th November 2014, Thursday
(3) 13th November 2014, Wednesday
(4) 12th November 2014, Tuesday
(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (1st Sitting) TF No. 6636838)

TYPE-III

1. The Chairman of the Selection Committee arrived at the Interview room for conducting an interview at 10 minutes to 12 : 30 hrs. He was earlier by twenty minutes than the other members of the board, who arrived late by 30 minutes. At what time were the interview scheduled?
(1) 12 : 10 (2) 12 : 20
(3) 12 : 30 (4) 12 : 40
(SSC Statistical Investigators Grade-IV Exam. 31.07.2005)
2. By looking in a mirror it appears that it is 6 : 30 in the clock. What is the real time?
(1) 6 : 30 (2) 5 : 30
(3) 6 : 00 (4) 5 : 00
(SSC Statistical Investigators Grade-IV Exam. 31.07.2005)
3. A clock gains five minutes every hour. What will be the angle traversed by the second hand in one minute?

- (1) 360° (2) 360.5°
(3) 390° (4) 380°
(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (First Sitting))
4. After 9'0 clock at what time between 9 p.m. and 10 p.m. will the hour and minute hands of a clock point in opposite direction?
(1) 15 minutes past 9
(2) 16 minutes past 9
(3) $16\frac{4}{11}$ minutes past 9
(4) $17\frac{1}{11}$ minutes past 9
(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (Second Sitting))
5. At what time are the hands of clocks together between 6 and 7?
(1) $32\frac{8}{11}$ minutes past 6
(2) $34\frac{8}{11}$ minutes past 6
(3) $30\frac{8}{11}$ minutes past 6
(4) $32\frac{5}{7}$ minutes past 6
(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (First Sitting))
6. In Ravi's clock shop, two clocks were brought for repairs. One clock has the cuckoo coming out every sixteen minutes, while the other one has the cuckoo coming out every eighteen minutes. Both cuckoos come out at 12.00 noon. When will they both come out together again?
(1) 2 : 06 PM (2) 2 : 08 PM
(3) 2 : 24 PM (4) 2 : 32 PM
(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (Second Sitting))
7. Kamala would like to complete all her home-work before 10.00 p.m. in order to watch an important TV programme. She has 40 minutes assignment in each of her five prepared subjects. What is the latest time at which she can start and still complete her home-work in the time for the programme?

- (1) 6 : 40 p.m. (2) 6 : 30 p.m.
(3) 7 : 10 p.m. (4) 7 : 20 p.m.

(SSC Combined Matric Level
(PRE) Exam. 21.05.2000
(1st Sitting) (East Zone)

- 8.** Reaching the place of meeting on Tuesday 15 minutes before 08 : 30 hours I found myself half an hour earlier than the man who was 40 minutes late. What was the scheduled time of the meeting?

- (1) 08 : 00 hours
(2) 08 : 05 hours
(3) 08 : 15 hours
(4) 08 : 45 hours

(SSC Combined Matric Level
(PRE) Exam. 21.05.2000
(1st Sitting) (East Zone)

- 9.** A clock only with only dots marking 3, 6, 9 and 12 position has been kept upside down in front of a mirror. A person reads the time in the reflection of the clock as 4.50. What is the actual time ?

- (1) 08 : 10 (2) 02 : 40
(3) 04 : 50 (4) 10 : 20

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (IInd Sitting)
(Eastern Zone, Guwahati)

- 10.** A clock only with dots marking 3, 6, 9 and 12 O'clock position has been kept upside down in front of a mirror. A person reads the time in the reflection of the clock as 10 : 20. What is the actual time?

- (1) 08 : 10 (2) 02 : 40
(3) 04 : 50 (4) 10 : 20

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (1st Sitting)
(North Zone, Delhi)

- 11.** A clock goes slow from midnight by 5 mts. at the end of the first hour, by 10 mts. at the end of the second hour, by 15 mts. at the end of the 3rd hour and so on. What will be the time by this clock after 6 hours?

- (1) 6 : 00 am (2) 5 : 30 am
(3) 6 : 30 am (4) 5 : 15 am

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (1st Sitting)
(North Zone, Delhi)

- 12.** A clock goes fast by one minute during the first hour, by two minutes at the end of the second hour, by 4 minutes at the end of

3rd hour, by eight minutes by the end of 4th hour, and so on. At the end of which hour, will it be fast by just over sixty minutes ?

- (1) Fifth (2) Sixth
(3) Seventh (4) Eighth

SSC Combined Matric Level (Pre) Exam.
05.05.2002 (IInd Sitting)
(North Zone Delhi)

- 13.** A clock with only dots marking 3, 6, 9 and 12 positions has been kept upside down in front of a mirror. A person reads the time in the reflection as 9.50. What is the actual time?

- (1) 2 : 15 (2) 8 : 40
(3) 8 : 50 (4) 4 : 15

SSC Combined Matric Level (Pre) Exam.
05.05.2002 (IInd Sitting)
(North Zone Delhi)

- 14.** A clock with only dots marking 3, 6, 9 and 12 positions has been upside down in front of a mirror. A person reads the time in the reflection as 6 : 10. The real time is

- (1) 06 : 50 (2) 12 : 40
(3) 12 : 20 (4) 6 : 10

SSC Combined Matric Level (Pre) Exam.
12.05.2002 (1st Sitting)

- 15.** Reaching the place of meeting 20 minutes before 8 : 50 hrs Satish found himself thirty minutes earlier than the man who came 40 minutes late. What was the scheduled time of the meeting?

- (1) 08 : 20 (2) 08 : 10
(3) 08 : 05 (4) 08 : 00

SSC Combined Matric Level (Pre) Exam.
12.05.2002 (IInd Sitting)

- 16.** A clock with only dot markings 3, 6, 9 and 12 positions has been kept upside down in front of a mirror. A person reads the time in the reflection of the clock as 12:30 the actual that will be

- (1) 12 O'clock (2) 12 : 30
(3) 6 O'clock (4) 03 : 45

SSC Combined Matric Level (Pre) Exam.
12.05.2002 (IInd Sitting)

- 17.** The bus for Chennai leaves every 30 minutes from a bus depot. The enquiry clerk told a passenger that the bus for Chennai left 10 minutes ago, and the next bus will leave at 10 : 30 a.m. What was the time when enquiry clerk told this ?

- (1) 10 : 20 a.m. (2) 10 : 10 a.m.
(3) 10 : 00 a.m. (4) 09 : 50 a.m.

(SSC Level Data Entry Operator &
LDC Exam.28.10.2012 (1st Sitting)

- 18.** A watch reads 7.30. If the minute hand points West, then in which direction will the hour hand point ?

- (1) North (2) North East
(3) North West (4) South East

(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)

- 19.** If 50 minutes ago, it was 45 minutes past four 'O clock, how many minutes is it until six 'O clock ?

- (1) 45 (2) 15
(3) 25 (4) 35

(SSC Constable (GD)

Exam. 04.10.2015, IInd Sitting)

- 20.** Ram leaves his house at 20 mins to seven in the morning, reaches Kunal's house in 25 mins. They finish their breakfast in another 15 mins and leave for their office which takes another 35 minutes. At what time do they leave Kunal's house to reach their office?

- (1) 7 : 40 A.M (2) 7 : 20 A.M
(3) 7 : 45 A.M (4) 8 : 15 A.M

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IInd Sitting)

TYPE-IV

- 1.** Which among the following is a Leap year ?

- (1) 1600 (2) 1900
(3) 1800 (4) All of the above

(SSC Level Data Entry Operator &
LDC Exam.21.10.2012 (IInd Sitting)

- 2.** Ram was born on 29th February. He celebrated his birthday falling on exactly 29th February 2008 for the fourth time. In which year he was born ?

- (1) 1992 (2) 2004
(3) 2000 (4) 1996

(SSC Graduate Level Tier-I

Exam.11.11.2012 (1st Sitting)

- 3.** Which of the following years did not have 29 days in February month?

- (1) 2000 (2) 2004
(3) 1996 (4) 1966

(SSC Multi-Tasking Staff
Exam. 24.03.2013, 1st Sitting)

ANSWERS

TYPE-I

1. (1)	2. (1)	3. (3)	4. (1)
5. (1)	6. (2)	7. (2)	8. (4)
9. (3)	10. (2)	11. (4)	12. (4)
13. (2)	14. (3)	15. (4)	16. (3)
17. (2)	18. (4)	19. (1)	20. (1)
21. (4)	22. (2)	23. (3)	24. (3)
25. (1)	26. (3)	27. (4)	28. (2)
29. (1)	30. (3)	31. (4)	32. (3)

TYPE-II

1. (1)	2. (2)	3. (1)	4. (2)
5. (3)	6. (4)	7. (2)	8. (3)
9. (3)	10. (2)	11. (2)	12. (1)
13. (4)	14. (3)	15. (3)	16. (1)

TYPE-III

1. (1)	2. (2)	3. (2)	4. (3)
5. (1)	6. (3)	7. (1)	8. (2)
9. (2)	10. (1)	11. (2)	12. (3)
13. (2)	14. (3)	15. (1)	16. (3)
17. (2)	18. (3)	19. (3)	20. (2)

TYPE-IV

1. (1)	2. (4)	3. (4)	
--------	--------	--------	--

EXPLANATIONS

TYPE-I

- (1) According to question,
9th $\rightarrow$ Saturday
Therefore, $9 - 7$
 $= 2$ nd $\rightarrow$ Saturday
 $\therefore$ 1st $\rightarrow$ Friday
- (1) Anil reached the place on Friday and he was three days earlier than the scheduled day.
Therefore, the scheduled day
 $=$ Friday $+ 3$ days $=$ Monday
If he had reached on Sunday then he would have earlier than one day.

- (3) The day before yesterday was Friday. Therefore, today is Sunday.
The day-after-tomorrow will be Tuesday
Tuesday $+ 3 =$ Saturday

- (1) Day before yesterday was Thursday.
Today is Saturday.
Tomorrow will be Sunday.

- (1) Day before yesterday was Sunday.
Therefore, today is Tuesday.
Day after tomorrow will be Thursday.
Thursday $+ 3 =$ Sunday

- (2) The day after tomorrow is Sunday.
Therefore, today is Friday.

The day on tomorrow's day before yesterday $=$ Friday $- 1 =$ Thursday

- (2) Shashikant was born on 29th September 1999.

15th August, 1999 was Sunday.
Days upto 29th September from 15 August

$16 + 29 = 45$ days $= 6$ weeks 3 old days

Sunday $+ 3 =$ Wednesday.

- (4) Friday $\rightarrow 2$ days earlier
Therefore, scheduled day $=$ Friday $+ 2 =$ Sunday, Sunday $+ 3 =$ Wednesday

Therefore, I would have been late by 4 days including Sunday.

- (3) Two weeks earlier than 23rd was also Sunday.

$$23 - 7 = 16$$

$$16 - 7 = 9$$

4 days earlier than 9 means 5th

9th $\Rightarrow$ Sunday

8th $\Rightarrow$ Saturday

7th $\Rightarrow$ Friday

6th $\Rightarrow$ Thursday

5th $\Rightarrow$ Wednesday

- (2) Tomorrow $\Rightarrow$ Friday

Friday $+ 3 =$ Monday

Monday $= 15$ th

Other Monday $\Rightarrow 22, 29$

Therefore, 30th June $=$ Tuesday

- (4) Two days after Tuesday will be Thursday.

- (4) Three days before today was Friday.

Today is Monday.

Therefore, day after tomorrow will be Wednesday.

- (2) Today is Tuesday.

Therefore, the day after tomorrow would be Thursday.

- (3) The day after tomorrow would be Friday.

Today is Wednesday.

The day before yesterday was Monday.

- (4) The day before yesterday was Tuesday

Today, it is Tuesday $+ 2 =$ Thursday

Tomorrow $\Rightarrow$ Friday

The day after tomorrow $\Rightarrow$ Saturday

- (3) Today $+ 3 =$ Tuesday

$\therefore$ Today $=$ Tuesday $- 3 =$ Saturday

Yesterday $=$ Saturday $- 1 =$ Friday

Friday $- 4 =$ Monday

- (2) Today is Saturday.

Tomorrow will be Sunday.

Sunday $+ 4 =$ Thursday.

- (4) Today is Monday.

Yesterday was Sunday.

Sunday $- 3 =$ Thursday.

- (1) Today is Wednesday $+ 4 =$ Sunday

Two days after tomorrow

$=$ Sunday $+ 3 =$ Wednesday

- (1) In a non-leap year the first and the last day would be the same.

- (4) Today is Saturday $- 2 =$ Thursday

Yesterday $\rightarrow$ Wednesday

Wednesday $- 3 =$ Sunday

- (2) Birthday of Kiran $=$ Monday $+ 2 =$ Wednesday

Shivratri $=$ Wednesday

The day after Shivratri $=$ Wednesday $+ 1 =$ Thursday

- (3) Today is Thursday $+ 2 =$ Saturday

Therefore, tomorrow will be Sunday.

- (3) Tomorrow will be Thursday.

Thursday $+ 3$ Days $=$ Sunday

- (1) Thursday $+ 2 =$ Saturday

- (3) Today is Thursday $+ 2 =$ Saturday

Therefore, tomorrow will be Sunday.

- (4) Today is Wednesday $+ 2 =$ Friday

Therefore, Day after tomorrow will be Sunday.

- 28.** (2) Today is Friday + 2, i.e., Sunday.
Tomorrow will be Monday. The Day after tomorrow will be Tuesday and two days after Tuesday will be Thursday.
- 29.** (1) $50 \text{ weeks} = 50 \times 7 = 350 \text{ days}$
Ann is 300 days older than Varun.
Sandeep is 350 days older than Ann.
Sandeep is $(350 + 300)$ days older than Varun.
Sandeep was born on Tuesday.
- $$650 \text{ days} = \frac{650}{7} = 92 \text{ weeks } 6 \text{ days}$$
- Number of odd days = 6
So, Varun was born 6 days after Tuesday, i.e., Monday.
- 30.** (3) Other Saturdays in March = 8, 15, 22, 29
Therefore, 1st April = Saturday + 3 = Tuesday
- 31.** (4) 27th September $\Rightarrow$ Saturday
30th September $\Rightarrow$ Tuesday
1st October $\Rightarrow$ Wednesday
Other Wednesdays $\Rightarrow$ 8, 15, 22 and 29
 $\therefore$ 27th October $\Rightarrow$ Wednesday - 2 = Monday
- 32.** (3) 3rd $\Rightarrow$ Tuesday
Other Tuesday $\Rightarrow$ 10th; 17th; 24th
 $\therefore$ 25th $\Rightarrow$ Wednesday

TYPE-II

- 1.** (1) 30th September 1998 $\Rightarrow$ Wednesday
30th September 1999 $\Rightarrow$ Thursday
30th September 2000 $\Rightarrow$ Saturday
Because 2000 is a Leap Year and there is one extra day in the month of February.
30th September 2001 $\Rightarrow$ Sunday
30th September 2002 $\Rightarrow$ Monday
30th September 2003 $\Rightarrow$ Tuesday
An ordinary year has one odd day.
- 2.** (2) 5 January 1965 $\Rightarrow$ Tuesday
5 January 1966 $\Rightarrow$ Wednesday
5 January 1967 $\Rightarrow$ Thursday
5 January 1968 $\Rightarrow$ Friday
5 January 1969 $\Rightarrow$ Sunday
Since, 1968 is a Leap Year.
5 January 1970 $\Rightarrow$ Monday
5 January 1971 $\Rightarrow$ Tuesday
- 3.** (1) The year 1996 was a Leap Year.
Number of days remaining in the 1996
 $= 366 - 26 = 340 \text{ days}$
 $= 48 \text{ weeks } 4 \text{ odd days}$
1997, 1998 and 1999 together have 3 odd days.
2000 was a Leap year
Days upto 15th August 2000
 $31 + 29 + 31 + 30 + 31 + 30 + 31 + 15 = 228 \text{ days}$
$$\frac{228}{7} = 32 \text{ weeks } 4 \text{ odd days}$$

Now, total number of odd days
 $= 4 + 3 + 4 = 11$
$$\frac{11}{7} = 1 \text{ week } 4 \text{ odd days}$$

15th August 2000 was 4 days beyond Friday, i.e., Tuesday.
- 4.** (2) According to Hari, his father's birthday may be on 14th or 15th June.
According to Hari's sister, their father's birthday may be on 15th, 16th or 17th June.
Common Date $\Rightarrow$ 15th June
- 5.** (3) Two days after Monday means Wednesday.
5 $\rightarrow$ Wednesday
 $5 + 7 = 12 \rightarrow$ Wednesday
 $12 + 7 = 19 \rightarrow$ Wednesday
Therefore, Tuesday will precede 19th that month.
- 6.** (4) The year 1989 was a normal year.
Days upto 15 August 1989 from August 15, 1988 = $16 + 30 + 31 + 30 + 31 + 31 + 28 + 31 + 30 + 31 + 30 + 31 + 15 = 365 = 52 \text{ weeks } 1 \text{ day}$
Therefore, Wednesday + 1 = Thursday
- 7.** (2) Number of Days in
January $\Rightarrow 31 - 25 = 6$
February $\Rightarrow 28$ (2006 is not a leap year)
- March $\Rightarrow 31$
April $\Rightarrow 30$
May $\Rightarrow 31$
June $\Rightarrow 30$
July $\Rightarrow 31$
August $\Rightarrow 31$
September $\Rightarrow 23$
Total $\Rightarrow 241 \text{ days}$
- 8.** (3) Last Monday was December 29, 1975.
Tuesday $\rightarrow$ December 30, 1975
Wednesday $\rightarrow$ December 31, 1975
Thursday $\rightarrow$ January 1, 1976
Friday $\rightarrow$ January 2, 1976
- 9.** (3) Third Friday = 16th
 $\therefore$ First Friday = 2nd
First Tuesday = 6th
 $\therefore$ Fourth Tuesday = 27th
- 10.** (2) Number of days from September 15, 2000 to September 15, 2001
 $= 365 + 1 = 366$
 $366 \div 7 = 2 \text{ odd days}$
 $\therefore$ September 15, 2001 $\Rightarrow$ Saturday
- 11.** (2) Mondays $\Rightarrow$ 1st, 8th, 15th, 22nd and 29th
23rd $\rightarrow$ Tuesday
24th $\rightarrow$ Wednesday
25th $\rightarrow$ Thursday
- 12.** (1) 4th = Saturday
Other Saturdays $\Rightarrow$ 11, 28, 25
Therefore, 27th $\Rightarrow$ Monday.
- 13.** (4) 2nd, 9th, 16th, 23rd and 30th $\Rightarrow$ Sunday
Therefore, 31st $\Rightarrow$ Monday
- 14.** (3) 2008 was a Leap year.
26th $\rightarrow$ Tuesday
27th $\rightarrow$ Wednesday
28th $\rightarrow$ Thursday
29th $\rightarrow$ Friday
1st $\rightarrow$ Saturday
2nd $\rightarrow$ Sunday
- 15.** (3) Mondays $\Rightarrow$ 8, 15, 22 and 29
Therefore, 30th $\Rightarrow$ Tuesday
- 16.** (1) Two days back $\Rightarrow$ 9th November 2014 - Sunday
Therefore, today $\Rightarrow$ 11th November 2014 - Tuesday
Tomorrow $\Rightarrow$ 12th November 2014 - Wednesday

TYPE-III

1. (1) The Chairman of the Selection Committee arrived at 12.20 hrs.
Other members arrived at 12.40 hrs
Scheduled time of Interview
= 12.10 hrs
2. (2) When it appears 6 : 30 in mirror, the real time would be 5 : 30.
3. (2) Each second-space equals 1° .
A clock gains five minutes every hour.

It means the clock gains $\frac{5}{60}$ minutes in one minute.

$$\frac{5}{60} \times 360 = 30$$

The second hand will traverse 360.5° in one minute.

4. (3) At 9 O'clock, the minute hand is $9 \times 5 = 45$ minute – spaces behind the hour hand. Therefore, the minute hand will have to gain $45 - 30 = 10$ minute space over the hour hand.

$\therefore$ Gain of 55 minute spaces equals 60 minutes.

$\therefore$ Gain of 15 minute spaces equals

$$= \frac{60}{55} \times 15 = \frac{180}{11} = 16 \frac{4}{11}$$

Therefore, hour and minute hands of a clock point in opposite direction after 9 O'clock at

$$16 \frac{4}{11} \text{ minutes past 9.}$$

5. (1) Hands of clock will be together at $32 \frac{8}{11}$ minutes past 6.

There are 30 minute spaces between hour and minute hand at 6 O' clock.

The minute hand gains 55 minutes in 60 minutes

$\therefore$ It will gain 30 minutes in

$$\frac{60}{55} \times 33 = 32 \frac{8}{11} \text{ minutes}$$

6. (3) LCM of 16 and 18
= $2 \times 8 \times 9 = 144$

Both Cuckoos will come out together again at

$$12 : 00 + 2 : 24 = 2 : 24 \text{ PM}$$

7. (1) $5 \times 40 = 200$ minutes
= 3 hours 20 minutes

Now,

$$10 : 00 - 03 : 20$$

$$= 06 : 40 \text{ PM}$$

8. (2) I reached at
 $08 : 30 - 00 : 15$

$$= 08 : 15 \text{ hours}$$

$$\text{I was } 40 - 30$$

$$= 10 \text{ minutes late}$$

Therefore, scheduled time

$$= 08 : 15 - 00 : 10$$

$$= 08 : 05 \text{ hours.}$$

9. (2) The actual time would be 2:40

10. (1) The actual time would be 8:10.

11. (2) Time after 6 hours after midnight = 6 am

Clock will go slow in 6 hours

$$= 6 \times 5 = 30 \text{ minutes}$$

$$\therefore \text{ Time shown by the clock}$$

$$= 6 : 00 - 0 : 30 \text{ minutes}$$

$$= 5 : 30 \text{ am.}$$

12. (3) First hour $\rightarrow$ 1 minute

Second hour $\rightarrow$ 2 minutes

Third hour $\rightarrow$ 4 minutes

Fourth hour $\rightarrow$ 8 minutes

Fifth hour $\rightarrow$ 16 minutes

Sixth hour $\rightarrow$ 32 minutes

Seventh hour $\rightarrow$ 64 minutes

13. (2) The actual time would be 8:40

14. (3) The real time is 12 : 20.

15. (1) Satish reached at

$$08 : 50 - 00 : 20 = 08:30$$

He was thirty minutes earlier than the man who came 40 minutes late. It implies that Satish was 10 minutes late.

$\therefore$ Scheduled time of the meeting

$$= 8 : 30 - 0 : 10 = 8 : 20$$

16. (3) The actual time will be 6 O'clock.

17. (2)

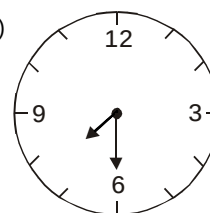
10:00 am

10m

$\rightarrow$ 10:10 am

10:30 am

18. (3)



The minute hand points West, it means the clock has been rotated through 90° clockwise. Therefore, hour hand will point North-West.

19. (3) Time at present

$$= 4 : 45 + 0 : 50 = 5 : 35$$

$$6 : 00 - 5 : 35 = 0:25$$

$$= 25 \text{ minutes}$$

20. (2) Ram left his house at 6.40 AM.

He reached Kunal's house at 7.05 AM.

They finished their breakfast in 15 minutes and left for office at 7.20 AM.

TYPE-IV

1. (1) The century which is completely divisible by 400 is a leap year.

$$= \frac{1600}{400} = 4$$

$$= \frac{1900}{400} = 4.75$$

$$= \frac{1800}{400} = 4.5$$

2. (4) Ram was born in a leap year. Ram celebrated his birthday for the third time in 2004.

Ram celebrated his birthday for the second time in 2000.

Therefore, he was born in the year 1996.

3. (4) A leap year is completely divisible by 4. In case of century years, only those divisible by 400 are leap years.

$$\frac{2000}{400} = 5 ; \frac{2004}{4} = 501 ;$$

$$\frac{1996}{4} = 499 ;$$

$$\text{But, } \frac{1966}{4} = 491.5$$